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Computational Methods in Combustion (MKWS)

Laminar Flame Speed and NO_x Formation in Ammonia–Hydrogen–Air Blends

Effect of hydrogen enrichment, equivalence ratio and preheat temperature

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1 Introduction

Ammonia (NH_3) has attracted strong renewed interest as a carbon-free fuel and as a practical hydrogen carrier: it contains no carbon, liquefies under mild pressure, and benefits from an existing global production and transport infrastructure. Its direct use in combustion systems is, however, hampered by two well-known drawbacks. First, ammonia burns very slowly — its laminar flame speed S_L at stoichiometric, atmospheric conditions is only about 6 cm s^{-1} , roughly one fifth of that of methane. This makes flame stabilisation difficult and narrows the operating envelope. Second, because ammonia carries fuel-bound nitrogen, it is prone to high nitrogen-oxide (NO_x) emissions.

A widely studied remedy is to blend ammonia with hydrogen (H_2). Hydrogen has an extremely high reactivity and flame speed, so even modest additions substantially accelerate the ammonia flame. The price of this improvement is typically a higher flame temperature and, with it, a tendency toward increased thermal NO_x . The fuel therefore presents an intrinsic *flame-speed versus NO_x trade-off* whose shape depends on the blend composition, the mixture stoichiometry and the operating temperature.

The aim of this project is to characterise that trade-off quantitatively using one-dimensional laminar flame simulations in Cantera. Three control variables are swept: the hydrogen mole fraction in the fuel X_{H_2} , the equivalence ratio φ , and the unburned-mixture (preheat) temperature T_{in} . The results are summarised as contour maps of S_L and NO_x , as an explicit Pareto front of the trade-off, and as an analysis of the preheat effect. The study identifies operating regions that combine an acceptable flame speed with a substantially reduced NO_x output.

2 Theoretical Background

2.1 Laminar flame speed

The laminar flame speed S_L is the velocity at which a planar, one-dimensional, premixed flame front propagates into a quiescent unburned mixture, measured relative to the unburned gas. It is a fundamental property of a fuel–oxidiser mixture that depends on chemical kinetics, thermodynamics and molecular transport. In a freely propagating flame the mass burning rate $\dot{m} = \rho_u S_L$ is an eigenvalue of the governing conservation equations and is obtained directly from the converged solution.

2.2 NO_x formation

Nitrogen oxides form through several coupled routes. In hydrocarbon and hydrogen flames the *thermal* (Zeldovich) mechanism dominates: it is strongly temperature-

activated and therefore grows rapidly once the flame temperature exceeds roughly 1800 K. In ammonia combustion an additional *fuel* NO_x route is important, because nitrogen released from the fuel is partly oxidised to NO via amine (NH_i) intermediates. The net NO_x therefore reflects a balance between formation (favoured by high temperature and excess oxygen) and reduction (favoured by fuel-rich conditions, where NO is consumed by NH_i radicals — the basis of the “NH₃-deNO_x” / SNCR effect). In this work NO_x is reported as the sum of NO and NO₂ mole fractions in the burned gas.

2.3 Equivalence ratio and preheat

The equivalence ratio φ measures the fuel-to-oxidiser ratio relative to stoichiometric: $\varphi < 1$ is lean, $\varphi > 1$ is rich. Both S_L and the flame temperature usually peak slightly rich of stoichiometric. Preheating the unburned mixture raises the flame temperature and accelerates the kinetics, increasing S_L markedly, but it also tends to raise thermal NO_x.

3 Numerical Model

3.1 Configuration

The simulations use the one-dimensional freely propagating premixed flame model **FreeFlame** of the open-source chemical-kinetics library Cantera (version 3.2). The flame is solved on an adaptively refined grid; mixture-averaged transport is used as a balanced compromise between accuracy and computational cost. All cases are computed at a pressure of 1 atm. Air is represented with the molar composition O₂:1, N₂:3.76.

The chemistry is described by the GRI-Mech 3.0 mechanism (`gri30.yaml`, 53 species, 325 reactions), which includes ammonia, hydrogen and a complete nitrogen sub-mechanism (NO, NO₂, N₂O). The fuel is a binary NH₃/H₂ blend characterised by the hydrogen mole fraction

$$X_{H_2} = \frac{n_{H_2}}{n_{H_2} + n_{NH_3}}. \quad (1)$$

3.2 Control variables

Three parameters are swept independently, giving a total of $4 \times 7 \times 3 = 84$ converged flames:

- Hydrogen fraction in fuel: $X_{H_2} \in \{0.0, 0.2, 0.4, 0.6\}$;
- Equivalence ratio: $\varphi \in \{0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3\}$;
- Preheat temperature: $T_{in} \in \{300, 450, 600\}$ K.

3.3 Solution strategy and post-processing

A *continuation* strategy is used: within each φ sweep, the converged solution of the previous case is supplied as the initial guess for the next. This significantly accelerates the sweep and improves robustness near the lean and rich flammability limits. For each converged flame the following scalars are extracted: the laminar flame speed S_L (the unburned-side flow velocity), the adiabatic flame temperature T_{ad} (burned-gas temperature), and the exhaust mole fractions of NO, NO₂ and N₂O. The quantity NO_x is defined as

$$\text{NO}_x = X_{NO} + X_{NO_2}, \quad (2)$$

expressed in parts per million (ppm). All results are written to a CSV file and processed with NumPy, pandas and Matplotlib.

4 Results and Discussion

4.1 Flame structure

Figure 1 shows the internal structure of a representative flame ($X_{H_2} = 0.4$, $\varphi = 1.0$, $T_{in} = 300$ K). The temperature rises sharply across a thin reaction zone while the fuel species NH₃ and H₂ are consumed; NO and the radical OH are produced in and just behind the front. This confirms that the model captures both the heat-release structure and the nitrogen chemistry needed for the present study.

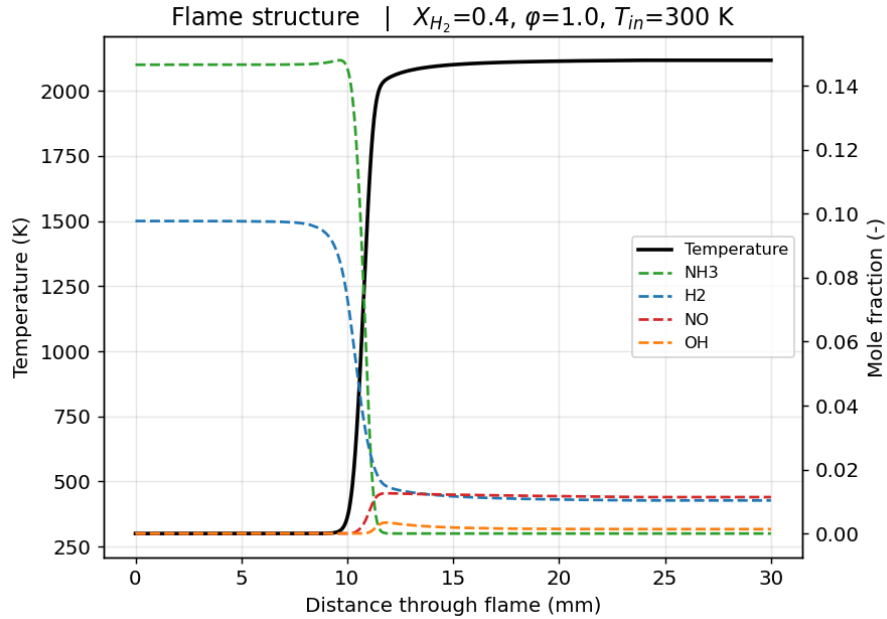


Figure 1: Resolved flame structure: temperature (left axis) and selected species mole fractions (right axis) through the flame, for $X_{H_2} = 0.4$, $\varphi = 1.0$, $T_{in} = 300$ K.

4.2 Effect of hydrogen enrichment on flame speed

Figure 2 shows the laminar flame speed as a function of X_{H_2} for several equivalence ratios at $T_{in} = 300$ K, and Figure 3 presents the same data as a contour map on the (φ, X_{H_2}) plane. Hydrogen addition has a strong, strongly non-linear effect. At stoichiometric conditions S_L rises from 5.9 cm s^{-1} for pure ammonia to 48.3 cm s^{-1} at $X_{H_2} = 0.6$ — an eight-fold increase. The flame speed peaks consistently near $\varphi \approx 1.1$, slightly rich of stoichiometric, as expected. Table 1 lists the peak flame speed for each blend.

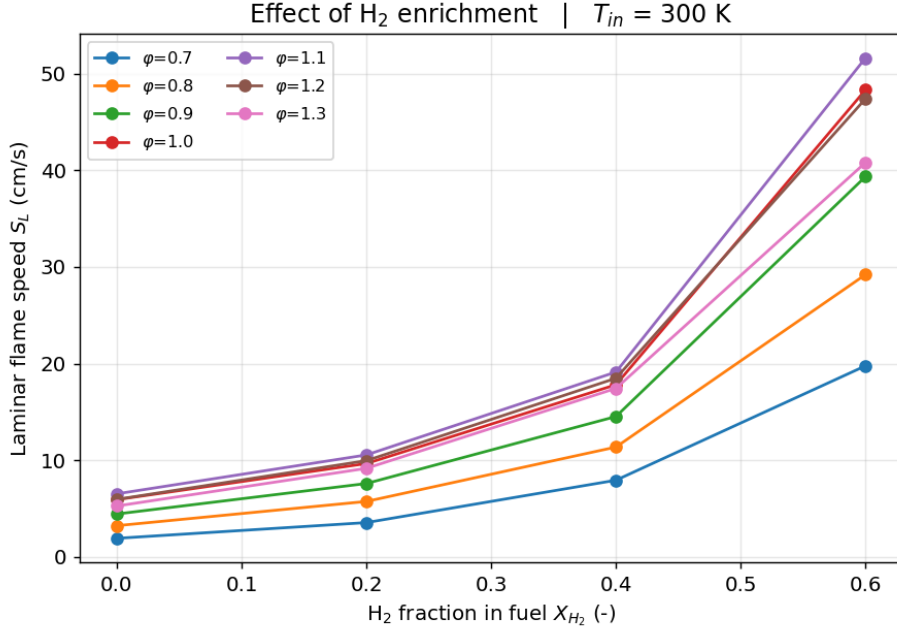


Figure 2: Laminar flame speed versus hydrogen fraction in the fuel, for several equivalence ratios ($T_{in} = 300$ K, $p = 1$ atm).

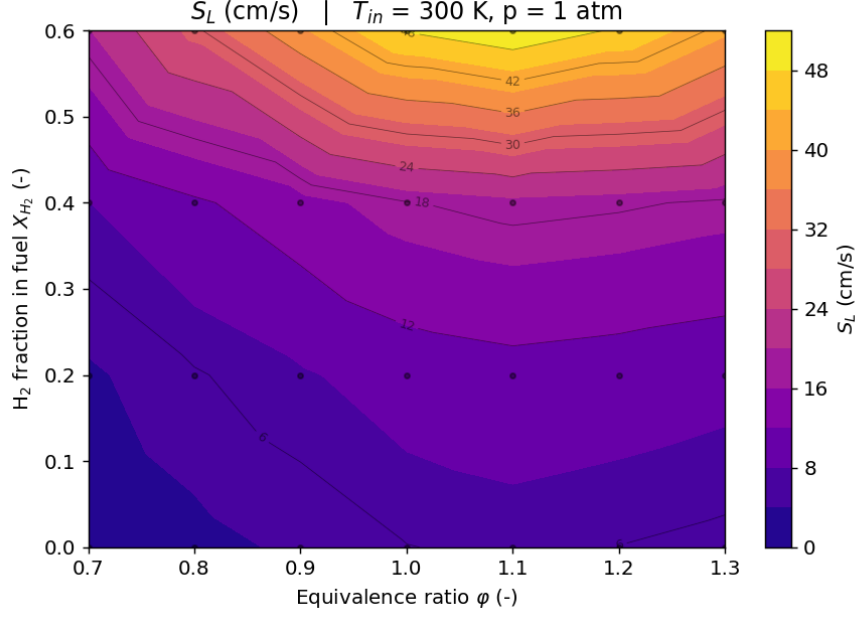


Figure 3: Contour map of laminar flame speed S_L (cm s^{-1}) on the (ϕ, X_{H_2}) plane at $T_{\text{in}} = 300$ K.

Table 1: Peak laminar flame speed and the corresponding NO_x for each blend ($T_{\text{in}} = 300$ K). The peak occurs at $\phi = 1.1$ in all cases.

X_{H_2}	S_L [cm s^{-1}]	T_{ad} [K]	NO_x [ppm]
0.0	6.5	2007	2084
0.2	10.6	2041	4676
0.4	19.1	2103	5354
0.6	51.6	2139	10391

4.3 NO_x emission

Figure 4 maps the exhaust NO_x over the same plane. Two trends stand out. First, NO_x peaks on the lean side, between $\phi \approx 0.8$ and 0.9 , where both flame temperature and oxygen availability are high; the peak shifts slightly leaner as hydrogen is added. Second, NO_x falls steeply as the mixture becomes rich: at $X_{H_2} = 0$ it drops from about 11 600 ppm at $\phi = 0.9$ to below 1400 ppm at $\phi = 1.2$, reflecting the consumption of NO by amine radicals under fuel-rich conditions. Increasing the hydrogen content raises NO_x overall, consistent with the higher flame temperatures it produces.

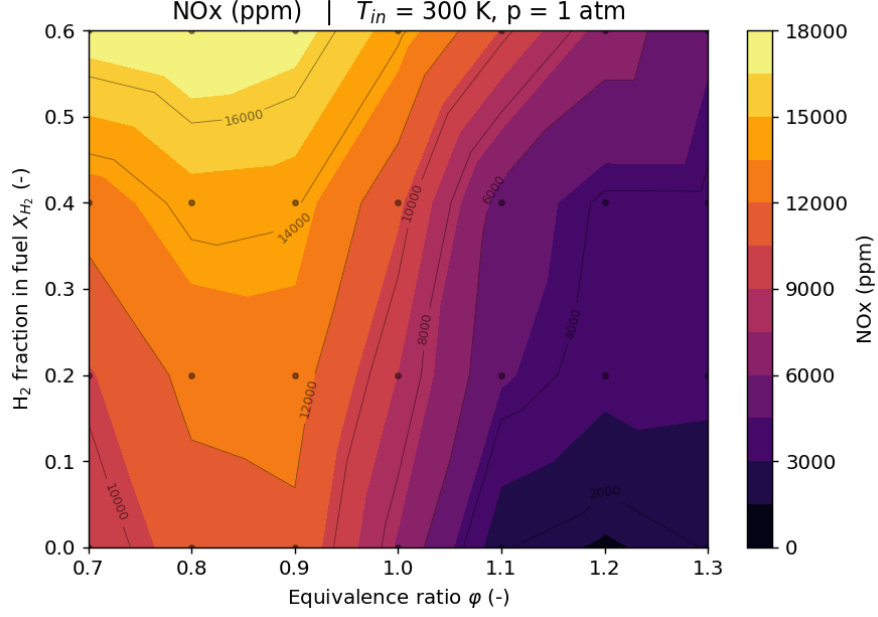


Figure 4: Contour map of exhaust NO_x (ppm) on the (ϕ, X_{H_2}) plane at $T_{\text{in}} = 300 \text{ K}$.

4.4 The flame-speed / NO_x trade-off

The competing influences of X_{H_2} and ϕ are summarised in the Pareto diagram of Figure 5, which plots NO_x against S_L with points coloured by equivalence ratio. The dashed Pareto front connects the operating points that are not dominated — those for which no other case offers both a higher flame speed and a lower NO_x . The front is populated almost entirely by fuel-rich points ($\phi \approx 1.2$ – 1.3 , the yellow markers): mildly rich operation retains most of the flame-speed benefit of hydrogen enrichment while cutting NO_x by a large margin relative to the lean and stoichiometric cases. This is the central practical result of the study.

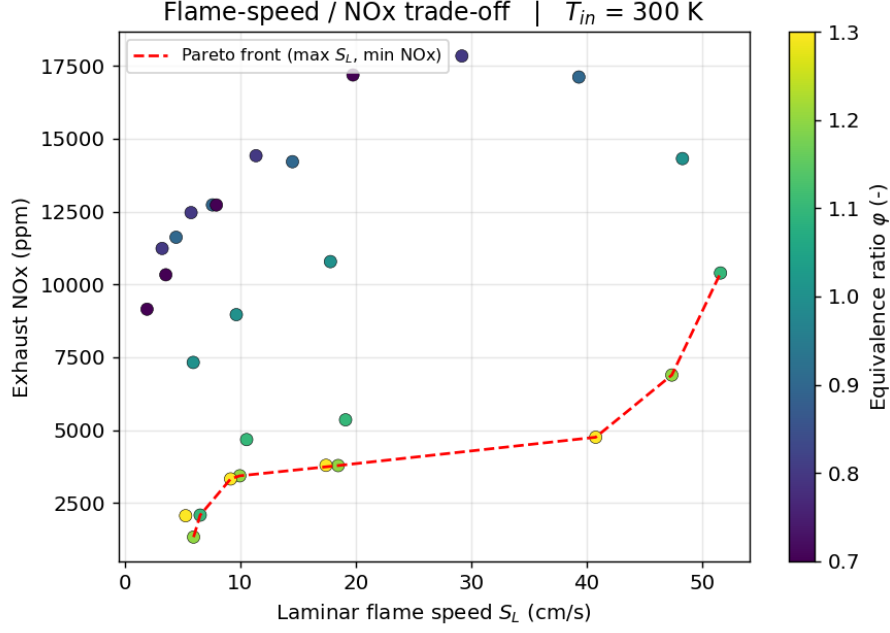


Figure 5: Flame-speed / NO_x trade-off at $T_{in} = 300$ K. Points are coloured by equivalence ratio; the dashed line is the Pareto front (maximum S_L , minimum NO_x).

4.5 Effect of preheat temperature

Figure 6 shows how preheating the unburned mixture affects S_L and NO_x at a fixed operating point ($X_{H_2} = 0.4$, $\phi = 1.0$). The effect on flame speed is dramatic: S_L grows from 17.8 cm s^{-1} at 300 K to 71.4 cm s^{-1} at 600 K, a roughly four-fold increase. The accompanying rise in NO_x is comparatively modest (from about 10 800 to 13 800 ppm), because the increase in flame temperature (≈ 170 K) is moderate. Preheating is therefore an effective lever for compensating ammonia's low reactivity at a limited NO_x penalty.

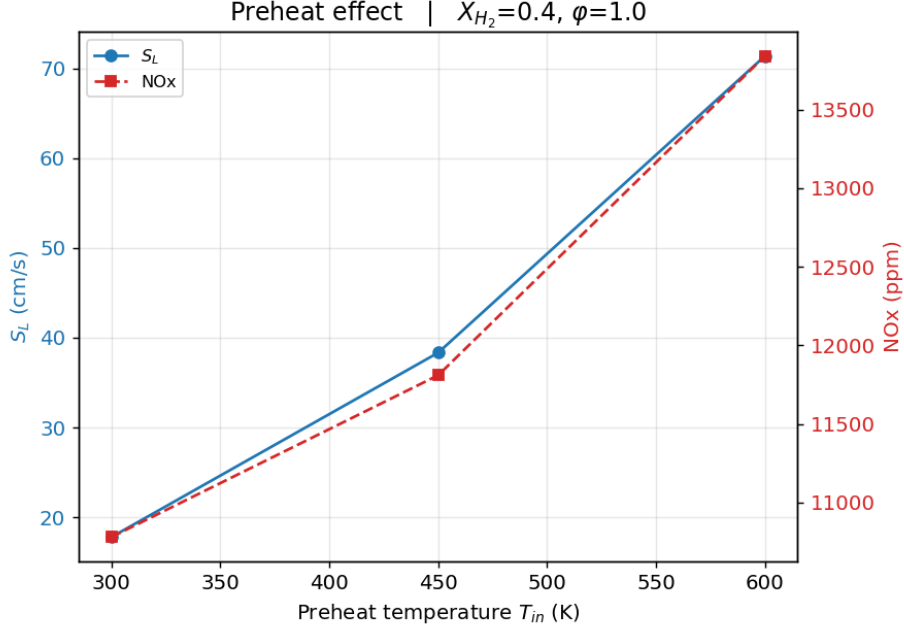


Figure 6: Effect of preheat temperature on laminar flame speed and exhaust NO_x ($X_{H_2} = 0.4$, $\phi = 1.0$, $p = 1$ atm).

5 Conclusions

A parametric Cantera study of one-dimensional laminar $\text{NH}_3/\text{H}_2/\text{air}$ flames was carried out, covering hydrogen fractions of 0–0.6, equivalence ratios of 0.7–1.3 and preheat temperatures of 300–600 K. The main findings are:

1. **Hydrogen enrichment strongly accelerates the flame.** At stoichiometric, atmospheric conditions the laminar flame speed rises from 5.9 cm s^{-1} (pure ammonia) to 48.3 cm s^{-1} at $X_{H_2} = 0.6$ — about an eight-fold increase. The peak flame speed occurs near $\phi = 1.1$ for every blend.
2. **NO_x peaks lean and collapses rich.** The highest NO_x occurs at $\phi \approx 0.8$ –0.9, while fuel-rich operation reduces NO_x by up to an order of magnitude through NO consumption by amine radicals.
3. **The best compromise is mildly rich operation.** The Pareto front of the flame-speed / NO_x trade-off is dominated by points at $\phi \approx 1.2$ –1.3, which keep most of the hydrogen-enrichment benefit while strongly cutting NO_x .
4. **Preheating is an efficient lever.** Raising T_{in} from 300 to 600 K quadruples S_L at only a moderate NO_x penalty.

The principal limitation is the use of GRI-Mech 3.0, which was optimised for natural-gas rather than ammonia chemistry; the absolute NO_x levels reported here

should be treated as indicative. A natural extension is to repeat the study with a dedicated NH_3/H_2 mechanism (e.g. Okafor or Stagni) and to compare the two, and to add a final high-fidelity run with multicomponent transport for the recommended operating points.

References

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